



CIS432

MACHINE LEARNING FOR BUSINESS ANALYTICS

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TEAM FINAL PROJECT

Optimizing Loan Application Decisions through Machine Learning
Design and Development of a Decision Support System for Efficient
Evaluation

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INTRODUCTION

Simon Bank of Rochester offers Home Equity Lines of Credit (HELOCs), allowing homeowners to borrow against their home equity. Currently, the review process for HELOC applications is conducted manually by expert loan officers, making it a time-consuming task. To improve efficiency, the bank is exploring automation for the initial screening and assessment of applications.

This report presents the development, results, and recommendations of a predictive machine learning model using decision trees, integrated into a decision support system (DSS) to automate the preliminary review of HELOC applications. The model addresses a binary classification problem:

1. Approve the loan
2. Deny the loan (with an explanation and guidance for future approval)

Using a dataset of approximately 10,000 past applications, the model was designed to assess eligibility accurately and provide actionable feedback for applicants who do not qualify.

This report details the methodology, evaluates the predictive performance of the model, and explores its integration into a decision support system that aims to reduce the workload of loan officers while ensuring human oversight remains where necessary.

The decision tree model used for binary classification achieved an overall accuracy of 68%, with an impressive 84% accuracy for negative predictions. Additionally, the model significantly reduced the workload of loan officers by nearly half, cutting it down by 47%. This reduction allows workers to redirect their efforts toward higher-value activities that contribute more strategically to the corporation. Furthermore, the decrease in workload can lead to reduced labor costs, enhancing overall operational efficiency.

METHODOLOGY: SELECTING A MODEL

The development of a predictive model and decision support system (DSS) for HELOC eligibility assessment required a solution that would satisfy four main requirements: easy automation, interpretability, accuracy, and regulatory compliance. After considering various machine learning models, a decision tree model was selected as the optimal solution to meet the requirements.

The evaluation process included considering different machine learning models. Logistic regression offered good interpretability and automation, but it's limited by its fundamental assumption of linear relationships between variables which made it inadequate for capturing the complex interactions inherent in financial data. Random forests demonstrated excellent accuracy and handling of non-linear relationships, but their complexity significantly reduced interpretability, limiting the requirement of offering actionable insights when a loan is denied.

A K-Means model was initially considered for classifying loans, but since it is an unsupervised clustering algorithm that groups data, it would be hard to explain to homeowners exactly why their

loan was denied. Gradient Boosting and XGBoost were also considered. A strong benefit of these models includes high predictive performance but since they work as black-box models, providing transparency is a big problem. Support Vector Machines possess good classification capabilities but are expensive and provide limited interpretability.

Decision trees emerged as the best option, meeting all critical requirements. In terms of automation, the model allows for real-time evaluation of applications and simplifies the initial screening process, greatly reducing the manual effort required by loan officers. The interpretability of decision trees is especially valuable, as the model builds a clear, rule-based framework that produces easy-to-understand decision paths. Decision trees generally have high accuracy indicators. Meaning the model performs well in classification tasks and reliably predicts denials.

Regulatory compliance is well-supported using the decision tree approach. The model maintains a transparent decision-making process and creates an auditable record of decisions, which is essential for meeting regulatory standards which is helpful when explaining denials.

Using decision trees can bring several operational benefits to the HELOC assessment process. The customer experience can significantly improve by providing instant automatic feedback. Applicants can receive clear explanations for decisions and, if needed, specific advice on how to strengthen future applications. From an operational point of view, the system can cut labor costs and improve resource allocation by allowing loan officers to focus on cases that require human judgment and add value to the company.

EXPLORATORY DATA ANALYSIS

After selecting the most fit model, an exploratory data analysis was conducted to fully understand data, the structure of the dataset, how many and which variables was the mode going to consider, identify patterns and trends, detect anomalies and outliers, treat missing values, duplicates look at the distribution and prepare the data for modeling. In this case, the original dataset had 10,459 entries and 24 variables. All of the variables are integers except for RiskPerformance which is the dependent variable.

Because data was missing from some of the entries, data preprocessing was necessary. There were 588 entries in which the entire row had missing values, in this case the entire row was deleted and the dataset ended up with 9871 rows with meaningful data. For certain missing values, it suffices to replace them with the average value of the variable. For others, missing data indicator columns were added to specify if and where each row is missing information. This is helpful because it allows to use an entry even if it is missing one or two values, which means it can be used to train the model instead of discarding it.

The above processing allowed the model for:

Preservation of Original Missing Information: The inclusion of missing indicators means that even after imputation, the model knows which values were originally missing. This can be useful if the pattern of missingness itself carries predictive information.

Preservation of Central Tendency: By replacing missing values with the mean, the overall average of the data in that column remains unchanged, ensuring that the central tendency is preserved.

THE MODEL

To provide intuitive explanations for the decision tree–based HELOC approval model, both the inherent interpretability of decision trees and supplemental techniques are utilized to clarify individual predictions. The decision tree structure follows a series of yes or no splits, where each node evaluates whether an applicant’s feature is above or below a specific threshold. This branching process allows for a straightforward visualization of how the model arrives at a “positive” (send to loan officer) or “negative” (deny) decision.

On a global level, the model’s feature importance and structure are analyzed to provide broader insights. In scikit-learn, for example, `model.feature_importances_` ranks variables from most to least predictive, while a tree plot (using `tree.plot_tree`) visually represents each branching criterion. By examining the tree from its root to its leaves, it becomes evident that features such as `ExternalRiskEstimate` and `AverageMinFile` appear often, indicating their significance in decision-making. This transparency allows the bank to understand which factors the model prioritizes and enables adjustments, such as implementing stricter thresholds for certain risk variables.

For individual predictions, particularly in cases where an applicant is denied, the model highlights the specific path the applicant’s data follows through the tree. Each internal node applies a simple rule, such as “Is the feature X below 0.3?” or “Is the feature Y above 600?” Tracing this path down to the final leaf explains exactly why a customer was classified as negative or positive. This approach ensures transparency in decision-making and provides applicants with clear reasons for their outcomes.

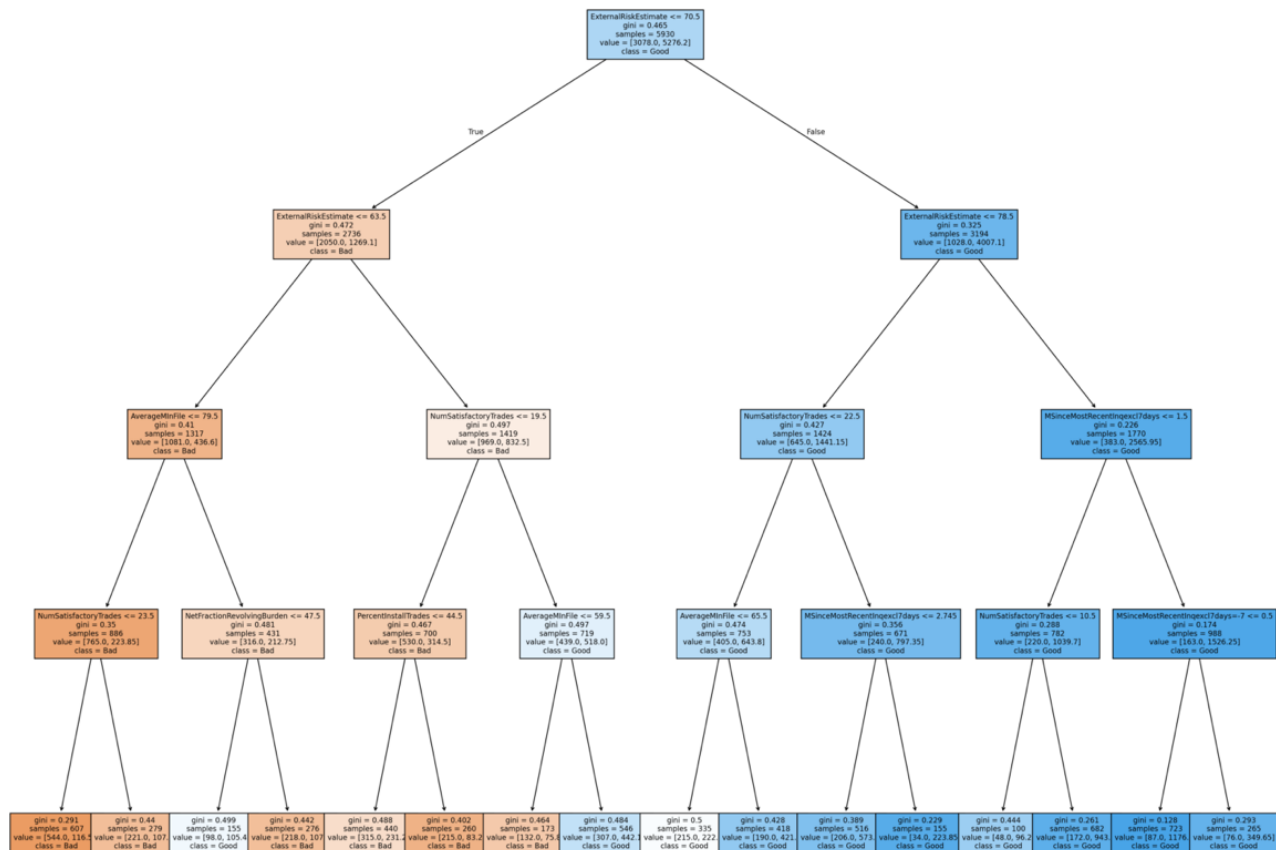


Figure 1: Decision Tree Model

STREAMLIT

A working prototype was built using Streamlit application, providing an interactive interface for users to explore the system's functionality. To enhance interpretability and user experience, Streamlit application only displays four key financial features out of the total 24 features available in the dataset i.e. Number Satisfactory Trades, Average Minimum File, External Risk Estimate and Net Fraction Revolving Burden. These features were identified as the most important for prediction based on the trained Decision Tree model. This approach simplifies user interaction by reducing the complexity of entering 24 different values. To make the app user-friendly and prevent unrealistic inputs, default values were set for each feature to the mean value for each feature. This ensures that users start with a reasonable baseline while still having the flexibility to adjust inputs for personalized predictions. By adjusting these inputs, users can see how the decision tree processes their information and determines eligibility, demonstrating the transparency and

interpretability of the model. Then, based on where the result ends up in the tree, user can assess why their loan was denied based on the data so they can strengthen their profile in the future.

STRATEGIES TO PREVENT PERFORMANCE DETERIORATION

For analysis, a decision tree was employed for model estimation. Decision trees are widely valued for their interpretability and ability to capture non-linear relationships.

However, the model's performance in a controlled analysis setting may not precisely reflect real-world outcomes in evolving scenarios over time.

1. **Periodic Model Retraining:** Regularly retrain the decision tree on the most recent data to capture latest trends and changes in the data distribution.
2. **Robust Validation Frameworks - Shadow Testing:** Deploy the updated decision tree model in parallel (shadow mode) with the current production system. This allows for performance comparisons without impacting live operations.
3. **Adaptive Decision Thresholding and Rule Adjustment.** Adjusting decision thresholds and rules in response to market shifts and regulatory changes can further enhance accuracy and compliance. This ensures that predictions stay aligned with current market trends, outputs consistently meet evolving regulatory standards. Adapting decision-making to changing risk profiles and market conditions keeps the model robust and relevant over time.
4. **Continuous Monitoring:**
 - a. **Real-Time Performance Metrics:** Establish a monitoring system to track key metrics (e.g., accuracy, precision, recall, and F1-score) continuously. This allows for the early detection of performance declines.
 - b. **Alert Mechanisms:** Set up automated alerts to notify the team when performance metrics fall below predefined thresholds.

CONCLUSION

The decision tree model achieved an accuracy of 84% for negative predictions, significantly reducing loan officers' workload by nearly half (47%) through automation.

Just as Simon Bank of Rochester improved efficiency, reduced costs, and enhanced the customer experience through machine learning, many other companies have leveraged similar technologies to optimize their operations. For example, Airbnb employs machine learning models for fraud prevention, market forecasting, and user behavior analysis. Their anti-party technology successfully blocked 74,000 high-risk bookings globally during New Year's Eve in 2024, demonstrating the power of predictive analytics in mitigating risks.

In conclusion, in today's rapidly evolving technological landscape, corporations must embrace advanced tools and automation processes to enhance decision-making, drive business growth, and improve the experiences of employees, vendors, and customers alike. Even in today's landscape, it remains very possible to use technology in an informative and ethical way.